TEACHER GUIDE

High school business

Break-even



A review lesson. Students run a car dealership in firms of four and review break-even on their own deals. Seat each firm together. Each firm answers the topic tasks together.

The lesson

45-minute setting

Setup	Round 1	Round 2	Round 3	Final questions

EVERY ROUND

Firm decisions
3 min

→

Deals
4½–5 min

→

Topic tasks
4 min

→

B2B deal
2½ min
from Round 2

LESSON SETTINGS

30 min | **45 min** | 60 min | 75 min

What students do

Every student buys and sells cars. In Firm decisions, each role has its own job.



CEO

Staff · Sabotage from Round 2



Finance

Capital & Loans · Investments



Sales & Marketing

Marketing



Operations

Seasonal inventory

Topic tasks

Each firm gets its own figures. The method is the same. Your Live screen shows each task and every firm's answer.

1 Cars to break even

Round 1

They take up to 8 cars.

Cars to break even?

Livery and setup, one-time	**$8,750**
Customer pays per car	**$9,750**
You pay per car	**$8,000**

ANSWER

5 cars

$9,750 − $8,000 = $1,750;
$8,750 ÷ $1,750 = 5

Why: Each car adds $1,750, so 5 cars pay the setup.

2 Lower break-even

Round 1

You want to lower your break-even point.

Which two changes would lower it?

- **A** Pay more for each car
- **B** Sell more cars
- **C** Pay higher rent
- **D** Raise the selling price
- **E** Cut the fixed costs

ANSWER

✓ Raise the selling price
✓ Cut the fixed costs

Why: More added per car or lower fixed costs; selling more does not move it.

3 Rank by break-even

Round 2

Three deals: fixed costs, and what each car adds (price minus cost).

Order by break-even, fewest cars first?

- **A** $4,000 fixed, $500 a car
- **B** $9,000 fixed, $1,500 a car
- **C** $6,000 fixed, $1,500 a car

ANSWER

1. $6,000 fixed, $1,500 a car
2. $9,000 fixed, $1,500 a car
3. $4,000 fixed, $500 a car

Why: Fixed costs ÷ what each car adds: 4, 6 and 8 cars.

Topic tasks

4 Profit on the deal

Round 2

A deal with a setup cost.

Profit on the deal?

Livery and setup, one-time	**$7,000**
Customer pays per car	**$10,750**
You pay per car	**$9,000**
They order	**9 cars**

ANSWER

$8,750

9 × ($10,750 − $9,000) − $7,000 = $8,750

Why: The first cars pay the setup; the rest is profit: $8,750.

5 Less risky deal

Round 3

Both deals are for 10 cars. The taxi deal breaks even at 4 cars, the rental deal at 8.

Which deal is less risky, and why?

A The taxi deal

B The rental deal

ANSWER

✔ **The taxi deal**

Why: The taxi deal's margin of safety is 6 cars; the rental deal's only 2.

6 New break-even

Round 3

They want $400 off a car.

New break-even in cars? Round up.

Livery and setup, one-time	**$5,500**
Customer pays per car now	**$19,250**
You pay per car	**$17,000**
Price cut per car	**$400**

ANSWER

3 cars

$18,850 − $17,000 = $1,850; $5,500 ÷ $1,850 → 3

Why: Less added per car (price minus cost) means more cars to break even.

Final questions

Each student answers 6 final questions. They come from these 10 questions on pricing.

1. An EV hatchback costs your firm $14,000 at Wholesale. You want a $3,000 gross profit. What price do you ask the customer, before sales tax?
   ✔ **$17,000**

2. You price a Family SUV at $19,000; it cost $16,000. A rival offers the same car at $18,500. If you match it, what is your gross margin as a % of your price?
   ✔ **13.5%**

3. A customer says your $21,000 price "is really $22,470". Why?
   ✔ **Indiana sales tax of 7% is added on top of your price at the sale**

4. You sell a car to another dealer, who gives you a resale certificate. How much sales tax do you add?
   ✔ **None: a dealer-to-dealer sale for resale is exempt**

5. A Quad bike costs $5,000 and you sell it for $7,000. What is the markup on cost?
   ✔ **40%**

6. Your last Pickup truck cost $20,000. A customer will pay $23,000; another dealer offers $22,000. Which sale makes more profit, and by how much?
   ✔ **The customer, by $1,000**

7. Wholesale discount: 2 cars −2%, 3–4 cars −4%, 5 or more −6%. You buy 3 Small vans listed at $8,000 each. What do you pay?
   ✔ **$23,040**

8. Which of your prices should you compare with a rival's advertised price?
   ✔ **Your own price before sales tax: prices are shown before tax, and every dealer adds the same 7%**

9. You cut an E-van from $21,000 to $20,000; it cost $18,000. What happens to your gross profit on the van?
   ✔ **It falls from $3,000 to $2,000**

10. You need a $6,000 gross profit from three Forklifts that cost $12,000 each. What single price per forklift does it?
    ✔ **$14,000**